

Spin as a Transformation Class of Rotating Wave Modes

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Abstract

This note examines the hypothesis that the wave function is a combination of rotating and non-rotating wave modes, spin expressing the rotation of these components rather than the rotation of a particle-like object. For the photon the hypothesis is nearly literal: the transverse components of a circularly polarized wave rotate, and the associated angular momentum is mechanically measurable. A scalar counterexample shows that the rotation of components is not sufficient: spin is fixed by the way the components transform under rotations of physical space. The resulting hierarchy is: the relative phase is a measurable coordinate within a sector, the weight m is an axial winding degree, the spin j labels the irreducible representation of $SU(2)$, and univalence $(-1)^{2j}$ separates superselection sectors as a rule on the algebra of observables. The formulation avoids the historical superluminal-velocity objection, which constrains rigid bodies rather than internal rotations of wave components. Any rotational ontology of spin must involve a relational, extended, or topologically attached wave configuration, never a point particle endowed with a mere internal axis.

Keywords: spin; helicity; spinor; $SU(2)$ representations; univalence; superselection; neutron interferometry; rotating wave modes; Riemann–Silberstein vector.

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1 Introduction

Spin is commonly introduced through negations: a rotation that is not a rotation, an angular momentum that is not orbital, a magnetic moment without a current loop. Faced with these negations, a natural hypothesis arises, and it is the founding hypothesis of this note: the wave function is a combination of rotating and non-rotating wave modes, and spin expresses the rotation of these components. The rotating entity is from the outset a wave, not a particle-like object endowed with an internal axis; the image is suggested by circularly polarized light, where two transverse field components in quadrature produce a genuinely rotating field vector. The task is then to determine which features of this rotation are physical, which are measurable, and what the hypothesis must still explain.

This note assesses that hypothesis systematically. Section 2 treats the photon, where the hypothesis is at its strongest. Section 3 states the hypothesis in its rigorous form and identifies the transformation law of the components under spatial rotations as the decisive criterion. Section 4 recalls why discrete spin values cannot be identified with particular phase angles, and locates the correct discrete structure in the representation theory of $SU(2)$. Section 5 presents the interferometric observability of the spinor sign. Section 6 formulates the univalence superselection rule as a statement about the algebra of observables. Section 7 derives the constraints that any internal-rotation ontology of spin must satisfy, and reviews the known constructions that meet them. Section 8 shows how the rotating-wave formulation dissolves the historical superluminal-velocity objection while distinguishing internal rotation from the separate causality problem raised by any literal spatial flow. Section 9 records the consequences for the Cosmochrony programme. Section 10 states the resulting structural hierarchy.

2 The photon: rotating fields and helicity

In a plane electromagnetic wave in vacuum the electric and magnetic fields are perpendicular in space but oscillate in phase: $\mathbf{B} = \hat{\mathbf{k}} \times \mathbf{E}/c$, and both vanish simultaneously. The rotation relevant to spin does not involve a phase shift between $\mathbf{E}$ and $\mathbf{B}$. It involves two transverse components of the electric field itself,

$$\mathbf{E}(z, t) = E_x \hat{\mathbf{x}} \cos \alpha + E_y \hat{\mathbf{y}} \cos(\alpha + \delta), \quad \alpha = kz - \omega t + \alpha_0. \quad (1)$$

For $\delta = \pm\pi/2$ and $E_x = E_y$ the field vector rotates about the propagation axis, and the sense of rotation corresponds to the helicity $\lambda = \pm 1$ of the photons, i.e. to the projection $S_{\parallel} = \pm\hbar$ of their spin on the propagation direction.

Two facts support the rotational intuition here. First, the angular momentum carried by circularly polarized light is mechanically real: a suspended birefringent plate experiences a measurable torque when it reverses the polarization of a light beam [3]. Second, the phase relation δ between the transverse components does determine the polarization state and hence the mean helicity.

Two facts, however, bound the intuition. Linear polarization ($\delta = 0$ or π) is not a zero-helicity photon state; it is a coherent superposition of the two helicities, and a helicity measurement yields $\pm\hbar$, never 0. The absence of a longitudinal (helicity-zero) mode for the free photon follows from masslessness and gauge invariance, through the finite-helicity representations of the massless little group $ISO(2)$, in which the translational part acts trivially [16]; Wigner’s classification also contains continuous-spin representations, and a massive spin-1 boson does possess three states. Moreover, a one-photon state of definite helicity has a vanishing mean electric field, $\langle \mathbf{E} \rangle = 0$, whereas a circularly polarized coherent state has a rotating mean field. The rotating classical field is therefore the coherent-state manifestation of the helicity structure, not a literal microscopic mechanism of photon spin.

3 Rotating waves rather than rotating objects

The founding hypothesis can now be stated precisely: the rotating entity is not a particle-like object but a wave mode, the wave function being a combination of rotating (chiral) and non-rotating components.

For the photon this language is physically literal. The state decomposes on the two circular polarization modes, $\Psi = \psi_+ |+\rangle + \psi_- |-\rangle$, which are the two helicity sectors, and the Maxwell fields can be packaged into the Riemann–Silberstein combinations

$$\mathbf{F}_\pm = \mathbf{E} \pm ic\mathbf{B}, \quad (2)$$

which separate the sectors of opposite helicity and support a genuine photon wave-function formalism [4]. For real Maxwell fields $\mathbf{F}_- = \overline{\mathbf{F}_+}$, so the two combinations are not independent real fields; the clean helicity separation is obtained after complexification and restriction to the positive-frequency components. Speaking of right- and left-rotating waves is then physically pertinent. A non-rotating wave, such as a linear polarization, is not a spin-zero state but a coherent superposition of the two counter-rotating modes; its mean angular momentum may vanish, yet a helicity measurement always yields $\pm\hbar$. The distinction $\langle S \rangle = 0$ versus $j = 0$ is essential: the former can result from the mutual cancellation of two opposite modes, the latter designates a wave that transforms as a genuine scalar under rotations.

A counterexample fixes the decisive criterion. A complex scalar wave $\psi = Ae^{-i\omega t}$ can be viewed as two real components in quadrature, $\psi = A \cos \omega t - iA \sin \omega t$, and thus “rotates” in its complex plane; yet its quanta can be of spin zero. The rotation of two components is therefore not sufficient: what matters is how the components transform under rotations of physical space,

$$\Psi(\mathbf{x}) \longmapsto D^{(j)}(R) \Psi(R^{-1}\mathbf{x}). \quad (3)$$

If $D(R) = 1$ the wave is scalar (spin 0); a vectorial $D(R)$ gives spin 1; a spinorial $D(R)$ changing sign after 2π gives spin $\frac{1}{2}$; higher representations give $j = \frac{3}{2}, 2, \dots$. The internal rotation counts only insofar as its plane is geometrically coupled to the directions of physical space; a rotation in a purely internal plane, as in the scalar counterexample, carries no spin.

The rotating-wave hypothesis can thus be stated rigorously: a one-particle state is represented by a carrier-valued wave amplitude; its spin is fixed by the carrier’s transformation law; its spin state indicates which combination of waves rotating in the various senses is realized when such a decomposition exists. For a massive fermion, however, chirality, helicity, and the weight m are distinct notions, and the counter-rotating decomposition is literal only in special cases such as photon polarization; in general the components form a carrier space whose transformation law, not its visual rotation, defines the spin. This is compatible with current physics, and it is native to the ground where the deeper constructions of Section 7 operate, since solitons, skyrmions, and Finkelstein–Rubinstein quantization concern extended field configurations rather than point objects. What remains to be derived is why the components of the wave carry precisely a given representation $D^{(j)}$: the rotation of the wave can represent the spin, but the topology and transformation law of the wave must still explain why it is scalar, vectorial, or spinorial.

Finally, this internal rotation must be distinguished from a vortex wave, $\psi(r, \varphi, z) \propto e^{i\ell\varphi}$, whose phase genuinely winds around a spatial axis and produces an orbital angular momentum $L_z = \ell\hbar$. A wave can carry both, $J_z = L_z + S_z$, with a spatial orbital winding and an internal polarization rotation. For the electromagnetic field the split of J into orbital and spin parts is not gauge invariant in full generality; it becomes operational in the paraxial regime, where spin and orbital angular momenta of Laguerre–Gaussian beams are separately measurable [1]. The rotating-wave intuition targets the spin part.

4 Weights, windings, and representations

The temptation is to associate the discrete spin values $j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ with particular phase angles such as $\delta = 0, \pi/2, \pi$ in Eq. (1). This fails, because δ varies continuously and parametrizes states *within* a given state space,

whereas j labels the space itself through its transformation law under all rotations.

A physical rotation acts on a spin- $\frac{1}{2}$ state by

$$U(\mathbf{n}, \alpha) = \exp\left(-\frac{i\alpha}{2} \mathbf{n} \cdot \boldsymbol{\sigma}\right), \quad (4)$$

so that $U(2\pi) = -I$ and only $U(4\pi) = +I$. The topology of the rotation group, $\pi_1(SO(3)) = \mathbb{Z}_2$, explains this dichotomy: it allows exactly two behaviours under a 2π rotation, $U(2\pi) = \pm I$, hence the integer and half-integer families. But the precise values of j are not elements of π_1 ; they arise from the theory of irreducible representations of $SU(2)$, where $2j$ is an integer highest weight, the degree of the representation $\text{Sym}^{2j}(\mathbb{C}^2)$.

A genuine winding number does survive after restriction to rotations about a fixed axis:

$$U_z(\alpha) |j, m\rangle = e^{-im\alpha} |j, m\rangle, \quad (5)$$

where $2m \in \mathbb{Z}$ is the winding degree of this $U(1)$ character over a full cycle of the covering group (4π). If m is half-integer, $e^{-im(2\pi)} = -1$ and $e^{-im(4\pi)} = +1$. The winding intuition is thus rigorous but axial and local: it describes the weights m , and j appears only when these weights are organized into a complete $SU(2)$ representation. It is also through the finite-helicity representations of the little group, together with the global choice of the relevant covering for half-integer values, that the helicity of massless particles is quantized [16].

For a spin- $\frac{1}{2}$ state written as

$$|\psi\rangle = \cos \frac{\theta}{2} |\uparrow\rangle + e^{i\phi} \sin \frac{\theta}{2} |\downarrow\rangle, \quad (6)$$

the continuous relative phase ϕ determines the orientation of the state on the Bloch sphere, not the intrinsic value of the spin: a measurement along a fixed axis yields only $\pm\hbar/2$.

5 Interferometric observability of the spinor sign

The sign reversal $U(2\pi) = -I$ is a global phase if the whole state is rotated, and as such is unobservable. It becomes observable when one branch of an interferometer is rotated relative to the other. Neutron interferometry realizes this: after a relative 2π spin rotation the two amplitudes interfere with opposite sign, and the initial interference pattern is restored only after 4π [10, 14].

What the experiment detects is not a hidden orientation of the neutron, but the spinorial character of the action of rotations on its quantum amplitude. The physically real structure is the transformation class of the state under rotations, since a state “rotated by 2π ” can be made to interfere with

itself and the difference observed. The rotation phase itself, by contrast, remains unobservable in isolation.

6 Univalence as a rule on the algebra of observables

Since $U(2\pi) = +1$ on integer-spin states and -1 on half-integer-spin states, a coherent superposition of the two would change its relative phase under a 2π rotation, that is, under no physical operation at all. The univalence superselection rule [15] draws the consequence.

The rule must be stated carefully. It does not forbid writing the vector sum $|\psi_B\rangle + |\psi_F\rangle$ of a bosonic and a fermionic state; such a vector exists mathematically. It states that no admissible physical observable measures its relative phase. Indeed, a 2π rotation maps

$$|\psi_B\rangle + e^{i\phi} |\psi_F\rangle \mapsto |\psi_B\rangle - e^{i\phi} |\psi_F\rangle, \quad (7)$$

and these two descriptions must be physically indistinguishable. The phase ϕ therefore enters no observable, and the would-be superposition is operationally equivalent to the corresponding mixture. In relativistic quantum field theory this becomes essentially the superselection of fermion parity.

Two kinds of phase must accordingly be distinguished. Within a single sector, as in neutron interferometry where both branches belong to the same half-integer sector, relative phases are measurable. Between the integer and half-integer sectors, no physical observable can serve as a coherent bridge, and the relative phase is suppressed by superselection. Univalence is thus a rule about the algebra of observables, not a mathematical impossibility of writing vector sums.

7 Constraints on internal-rotation ontologies

The preceding results define a precise specification for any interpretation of spin as an underlying rotation. Such a construction should not merely exhibit something that rotates. It should explain why $R(2\pi) \neq I$ while $R(4\pi) = I$ on the underlying object, render the first effect observable only by relative comparison, and reproduce the measurement probabilities, relativistic covariance, and the integer versus half-integer dichotomy.

A simple rotating arrow fails this test: an arrow returns to itself after 2π . The minimal classical configuration with the required topology is exhibited by Dirac's belt trick: a frame attached to its environment, whose configuration retains the relation to its surroundings or to the history of its rotation. A 2π rotation of such a frame cannot be undone with the attachments held fixed, whereas a 4π rotation can. This is the "orientation-entanglement" of Misner, Thorne, and Wheeler [8].

Orientation-entanglement is not, however, the only construction available. Extended objects can acquire fermionic behaviour upon quantization: solitons and skyrmions [11] under the Finkelstein–Rubinstein quantization conditions [5], and gravitational geons in the same framework. Their common feature confirms the structural lesson: the spinorial sign never comes from a local arrow, but from the global topology of the configuration space.

Closest to the initial intuition, Hestenes represents the Dirac spinor as a rotor of the Clifford algebra, encoding a local frame, a phase, and a rotation plane, in connection with the *zitterbewegung* of Dirac theory [7]. This is a substantial and illuminating geometric reformulation of the Dirac formalism, but it does not by itself derive spin $\frac{1}{2}$ from a more fundamental substructure.

The minimal ontological hypothesis compatible with all constraints is therefore the following: if spin originates in an underlying rotation, the rotating entity must be a relational, extended, or topologically attached wave configuration, never a point particle endowed with a mere internal axis. The rotating-wave formulation of Section 3 meets the negative part of this requirement, since it discards the rigid point-like object; but it does not by itself derive $U(2\pi) = -I$, the spinorial topology, or the quantum measurement probabilities, which remain precisely the results a deeper construction would have to produce.

8 Causality and the superluminal-velocity objection

Any rotational model of the electron faces a famous objection: if a sphere of classical electron radius carries an angular momentum $\hbar/2$, its equatorial velocity must exceed the speed of light by orders of magnitude. The difficulty was already noted by Uhlenbeck and Goudsmit themselves [13], formulated by Lorentz, and later pressed by Pauli against any material interpretation of spin; the detailed history is given by Giulini [6] (see also [12]). The objection targets a specific model: matter distributed at a spatial radius r , rotating mechanically at angular frequency ω , with local velocity $v = \omega r$. Three quantities must therefore be kept distinct: the angular frequency of an internal evolution, the phase velocity of a wave pattern, and the tangential velocity of a material distribution. Only the third is directly constrained by relativity.

The standard formalism, moreover, assigns the electron no intrinsic mechanical rotation frequency at all. A stationary state of definite spin evolves as $|\psi(t)\rangle = e^{-iEt/\hbar} |\mathbf{p}, s\rangle$: the global phase rotates at frequency $E/\hbar$ while $\langle \mathbf{S} \rangle$ remains constant. Spin precession occurs only in an external magnetic field, at the Larmor frequency $\boldsymbol{\Omega}_L \simeq -(gq/2m) \mathbf{B}$, which describes the evolution of the quantum orientation, not the rotation of a material surface.

For the rotating-wave hypothesis of Section 3 the objection dissolves. A

rotation in the internal space of the components, $\Psi \mapsto U(t) \Psi$, has for “radius” a field amplitude, not a distance; the product ωr is then not a spatial velocity. Circular polarization illustrates this: at a fixed point the direction of $\mathbf{E}$ rotates at the optical frequency, yet no material surface traverses a circle, and in an ideal plane wave the energy flux remains directed along the propagation.

Causality nevertheless remains binding wherever a literal circulation is asserted. If a density of charge or energy really circulates around a spatial axis, its local velocity must respect relativity; for the electromagnetic field the energy flux obeys $|\mathbf{S}_P| \leq c u$, with $\mathbf{S}_P$ the Poynting vector and u the energy density. A phase pattern may travel faster than c without transporting information or energy at that speed, but a real flow may not.

The wave formulation admits a field-theoretic reading of this constraint. Following Belinfante [2], Ohanian showed that the Dirac angular momentum can be expressed in terms of a circulating energy flow in the wave field, extended over the Compton scale $\lambda_C = \hbar/mc$ [9]. This removes the need for a surface velocity ωr at any assigned radius: the classical paradox arose from confining the rotation of a rigid body to the classical electron radius $r_e = \alpha \lambda_C \ll \lambda_C$. The result must not be over-read, however. It establishes an equivalent field-theoretic bookkeeping of the angular momentum as a flow, not literal circular trajectories with a unique local velocity, and the electromagnetic bound $|\mathbf{S}_P| \leq c u$ does not transpose automatically to the Dirac energy–momentum tensor. A literal hydrodynamic interpretation of this circulation would therefore require a separate causal analysis.

The verdict is thus twofold. An internal rotation of the wave components raises no superluminal problem at all. A spatial-circulation reading is available at the level of the field’s energy flow on the Compton scale, but its literal hydrodynamic interpretation is not established by the formalism, and the rotating-wave hypothesis must refrain from reintroducing, surreptitiously, a mechanical radius of the electron.

9 Implications for the Cosmochrony programme

The preceding structure sorts the current state of the Cosmochrony construction [?] into established facts, one exact consequence, one needed correction, one diagnostic, and one open question.

Established: the admissibility construction selects a two-dimensional complex module $V_\rho \simeq \mathbb{C}^2$, with $\text{Sym}^2(V_\rho) = V_1$ and $\wedge^2(V_\rho) = V_0$, reproducing the Clebsch–Gordan content $\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$.

Exact algebraic consequence: the central element $-I$ acts on $V_\rho^{\otimes n}$ as $(-1)^n$. Conditional on the Lorentz-spin identification of V_ρ , this tensor grading becomes the physical univalence grading of Section 6; absent that identification, it remains a grading of the admissible $SU(2)$ only.

Correction: in the Q7 construction [?], the state e_0 must be identified as the $m = 0$ weight of the $j = 1$ module, not as a $j = 0$ state; the genuine $j = 0$ state is the antisymmetric singlet in $\wedge^2(V_\rho)$.

Diagnostic: Lorentzian spin and the weak doublet must act on distinct tensor factors; a single copy of V_ρ cannot carry both actions without conflating spacetime and internal symmetries.

Open questions: two distinct problems must be separated. The representational problem is to construct the full action $\rho_{\text{rot}}: \text{Spin}(3) \rightarrow U(\mathcal{H}_\Pi)$ with $\rho_{\text{rot}}(-I) = (-1)^n I$ on the tensor sector of degree n ; a one-parameter evolution cannot supply the three non-commuting generators of $\text{Spin}(3)$, so the cascade alone cannot solve this problem. The dynamical problem, stronger, is to determine whether the cascade canonically traces the orbit of an axial one-parameter subgroup, $U_\Pi(\tau) = \rho_{\text{rot}}(\exp[-i\alpha(\tau)J_{\mathbf{n}}])$, with both the axis $\mathbf{n}$ and the relation $\alpha(\tau)$ derived rather than presupposed. Solving the first would be a necessary step toward a rotational derivation of spin; solving the second would tie that representation to the cascade dynamics; neither is yet a derivation of the full quantum formalism.

10 Conclusion

The founding hypothesis, spin as the rotation of the wave modes composing the wave function, survives in a corrected and stratified form:

relative phase : measurable coordinate within a sector, m : winding weight of the rotation about an axis, j : global representation of $SU(2)$, $(-1)^{2j}$: univalence separating the physical sectors.	(8)
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Spin is not the instantaneous angle of an internal rotation; it characterizes the global way the wave mode transforms under rotations, as expressed by Eq. (3). A one-particle state is represented by a carrier-valued wave amplitude, and its spin is fixed by the carrier's transformation law; only in special cases, notably photon polarization, can its components be interpreted literally as counter-rotating modes. For the photon this structure manifests itself classically as the chirality of polarization, measurable as mechanical angular momentum [3]. For fermions it requires a spinorial structure whose relative phases have measurable interferometric consequences [10, 14], and whose sign structure partitions the Hilbert space into superselection sectors [15]. The hypothesis escapes the superluminal-velocity objection, which constrains rigid bodies, not internal rotations of wave components [9]. The hierarchy of Eq. (8) is thus an experimentally anchored structure fixing the specification that any deeper rotational ontology of spin must meet.

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